

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0587

Mass 138 kV
March 17, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0587. The Interconnection Customer proposes a 63.8 MW solar photovoltaic facility interconnecting at the Mass 138 kV substation in the Kentucky area. The request was received on February 29, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 302 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,194,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Appleton - Leslie 138kV	Wixom - Albion Rd 138kV	162.5	224	33.0
Leslie - St Joseph 138kV L	Wixom - Albion Road 138kV	162.1	221	33.0
Davisville - Wellington 138kV	Albion Rd - Wixom 138kV	139.3	197	33.0
Ayers Island - Wixom 138kV L	Barnabie Cable Sta - Casino 138kV	138.9	257	40.8
Ubc South Campus - Danz Ave Sw Sta 138kV	Albion Road Substation - Platte City 138kV	131.6	192	23.1
Anson N - 51St St 138kV L	Wixom - Albion Road 138kV line	127.8	227	20.5
51St Street - Culver 138kV	Wixom - Albion Road 138kV	126.1	214	20.5
Campbell Jct - School Street Substation 138kV	Glendale South Tap 6 - Campbell Jct 138kV line	123.8	212	10.0
Danz Ave Sw Sta - Cwec Tigerton 138kV	Albion Rd - Platte City 138kV	121.0	202	23.1
Sand Springs - Ojo 138kV line	Sand Springs - Snohomish 138kV	120.8	232	17.9
Meeker - Sidney Coal 138kV line	Spring Crk Rd - Harris 138kV	119.2	231	7.3
Meeker - Turkey Crk 138kV	Spring Creek Rd - Harris 138kV	118.9	219	7.3
Chestnut Substation - Saint Joseph 138kV	Wixom - Albion Rd 138kV	116.7	295	33.0
Davisville - N Appleton 138kV	Wixom - Albion Rd 138kV line	113.6	329	33.0
Cwec Tigerton - Ellendale 138kV	Albion Rd - Plate City 138kV L	113.4	209	23.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hazard - Harris 138kV L	Sidney Coal - Scott Branch Substation 138kV Line	113.2	227	8.7
Tom Watkins - Turkey Crk 138kV L	Spring Crk Rd - Harris 138kV line	111.3	228	7.3
Spring Crk Rd Substation - Harris 138kV	Scott Branch Substation - Sidney Coal 138kV Line	110.6	315	12.8
Hotel Road - Silver Bay Power 138kV Line	Livermore - Lee County Solid Waste Energy 345kV	109.2	179	5.2
North Hendrickson - Clubhouse 138kV	Glendale S Tap 6 - Campbell Jct 138kV L	105.2	216	10.0
Greenville Ave - Beanblossom-Sw Sta 138kV	Lee County Solid Waste Energy - Livermore 345kV	105.1	284	8.3
Snohomish - Sand Springs 138kV	Snd Springs - Ojo 138kV	104.2	262	15.4
Anson N - Pin Hook Substation 138kV	Barnabie Cable Sta - Casino 138kV	103.7	395	34.8
Skeena - Pin Hook 138kV	Barnabie Cable Sta - Casno 138kV L	103.2	407	34.8
Campbell Jct - Rapids Energy Ctr 138kV	Cedar Tap - Terry Co 138kV	102.0	561	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Leslie - N Appleton 138kV	Rebuild line with higher-capacity conductor	\$4,530,000	6% (\$289,000)
Saint Joseph - Leslie 138kV L	Rebuild line with higher-capacity conductor	\$21,962,000	6% (\$1,403,000)
Wellington - Davisville 138kV	Rebuild line with higher-capacity conductor	\$4,486,000	10% (\$446,000)
Ayers Island - Wixom 138kV L	Rebuild line with higher-capacity conductor	\$14,086,000	12% (\$1,697,000)
Ubc S Campus - Danz Ave Sw Sta 138kV	Rebuild line with higher-capacity conductor	\$9,343,000	18% (\$1,672,000)
Anson N - 51St St 138kV line	Rebuild line with higher-capacity conductor	\$24,165,000	12% (\$2,797,000)
51St Street - Culver 138kV	Rebuild line with higher-capacity conductor	\$5,238,000	12% (\$606,000)
Campbell Jct - School Street Substation 138kV	Rebuild line with higher-capacity conductor	\$8,905,000	4% (\$323,000)
Danz Ave Sw Sta - Cwec Tigerton 138kV	Rebuild line with higher-capacity conductor	\$6,544,000	18% (\$1,171,000)
Ojo - Sand Springs 138kV	Rebuild line with higher-capacity conductor	\$13,529,000	9% (\$1,245,000)
Meeker - Sidney Coal 138kV	Rebuild line with higher-capacity conductor	\$5,748,000	1% (\$77,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Turkey Crk - Meeker 138kV	Rebuild line with higher-capacity conductor	\$404,000	1% (\$5,000)
Chestnut Substation - Saint Joseph 138kV	Rebuild line with higher-capacity conductor	\$4,642,000	6% (\$296,000)
Davisville - N Apleton 138kV	Rebuild line with higher-capacity conductor	\$5,970,000	6% (\$381,000)
Cwec Tigerton - Ellendale 138kV	Rebuild line with higher-capacity conductor	\$3,694,000	18% (\$661,000)
Hazard - Harris 138kV L	Rebuild line with higher-capacity conductor	\$32,433,000	3% (\$975,000)
Tom Watkins - Turkey Crk 138kV Line	Rebuild line with higher-capacity conductor	\$9,372,000	1% (\$125,000)
Spring Crk Rd Substation - Harris 138kV	Rebuild line with higher-capacity conductor	\$3,501,000	2% (\$73,000)
Hotel Road - Silver Bay Power 138kV Line	Reconductor line	\$7,592,000	4% (\$270,000)
North Hendrickson - Clubhouse 138kV	Reconductor line	\$1,324,000	4% (\$48,000)
Greenville Ave - Beanblossom-Sw Sta 138kV	Reconductor line	\$6,366,000	4% (\$230,000)
Snohomish - Sand Springs 138kV	Reconductor line	\$9,794,000	8% (\$827,000)
Anson North - Pin Hook 138kV	Reconductor line	\$5,284,000	8% (\$428,000)
Skeena - Pin Hook 138kV	Reconductor line	\$1,614,000	8% (\$131,000)
Campbell Junction Substation - Rapids Energy Ctr 138kV line	Reconductor line	\$1,318,000	1% (\$18,000)
	Total allocated to project		\$16,194,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.